

Linux System Utilities & Event Logging Project

Project Title: Keylogging & Automated Log Exfiltration

Total Marks: 10

Allowed Languages: Bash Scripting

Target OS: Ubuntu 22.04 LTS

1. Project Overview & Context

In Linux systems, hardware input devices (keyboard, mouse) interact with the kernel by generating input event logs managed via the virtual filesystem under `/dev/input/event*`. Understanding how raw hardware events are captured, parsed, audited, and periodically exfiltrated via email protocols (SMTP) is fundamental for system administration, forensic logging, and security monitoring.

Your objective is to build a lightweight, robust Linux utility script (in Bash or Python) that intercepts, logs, and formats keystroke input from a designated Linux input device file in real-time, and automatically emails the recorded log files to a specified email address at regular time intervals.

2. Project Requirements & Phase Specifications

Phase 1: Input Device Identification & Privileges (10 Marks)

- **Device Detection:** Automatically identify or allow the administrator to specify the active keyboard input file (e.g., inspecting `/proc/bus/input/devices` or parsing `/dev/input/by-id/`).
- **Privilege Verification:** Reading raw hardware event streams requires administrative access. The script must check if it is being executed with root privileges (`UID == 0` or `sudo`) and exit gracefully with an informative error if privileges are insufficient.

Phase 2: Real-time Event Capture & Logging (15 Marks)

- **Raw Event Interception:** Intercept and decode binary event packets from the designated `/dev/input/event*` interface.
- **Key-Mapping & Translation:** Map raw Linux input key codes (e.g., event type `EV_KEY`) to readable human text (handling special keys like `ENTER`, `SPACE`, `BACKSPACE`, `SHIFT`).
- **Timestamped Log Writing:** Store all captured keystrokes into a local log file named `keystroke_audit.log` with precise ISO timestamps.

Phase 3: Automated Email Exfiltration / Dispatch

- **Periodic Transmission Engine:** Implement an automated background timer mechanism (or integration with `cron`) to send the log file to a specified recipient email at configured intervals (e.g., every 5 minutes or 1 hour).

- **SMTP Integration:** Configure secure email sending using SMTP (e.g., Python smtplib / email modules or mailx/ssmtp in Bash)

3. Submission Deliverables

Submit a compressed archive named [StudentID]_Linux_Keylogger_Project.zip containing:

1. **keylogger.py / keylogger.sh:** Executable script code containing event capture and automated email sending logic.
2. **test_run.log & Screenshot:** Sample log file and screenshot proof of receiving the automated log email in the target inbox.